Experiment-05:

Name of the Experiment: Arduino with Simulation Software. (Blinking LED Traffic lights)

Software Used: TinkerCad

Components Required :

1. Resistor(220ohms)
2. Arduino Uno R3.
3. Three LED’s. (RGB)

Procedure

1. Go to TinkerCad.com for performing this simulation.
2. Go to components and select LED and Resistances.
3. Do the simulation according to the instructor.

Simulation: For this procedure a code is needed to be uploaded into Arduino Uno R3.

CODE:

```
void setup() {
pinMode(4, OUTPUT);
pinMode(3, OUTPUT);
pinMode(2, OUTPUT);
}
void loop() {
digitalWrite(4, HIGH); // For green LED
delay(3000); // Wait for 3 second
digitalWrite(4, LOW);
digitalWrite(3, HIGH); // For orange LED delay(1000); // Wait for 1 second
digitalWrite(3, LOW);
digitalWrite(2, HIGH); // For the red LED delay(3000); // Wait for 3 second
digitalWrite(2, LOW);
}
```

Simulation Output:

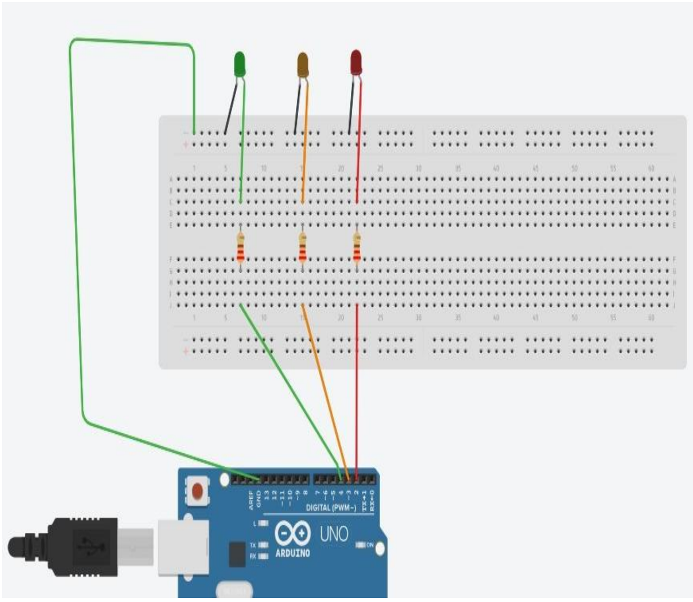

Fig 01 : Circuit on Tinkercad (Normal state)

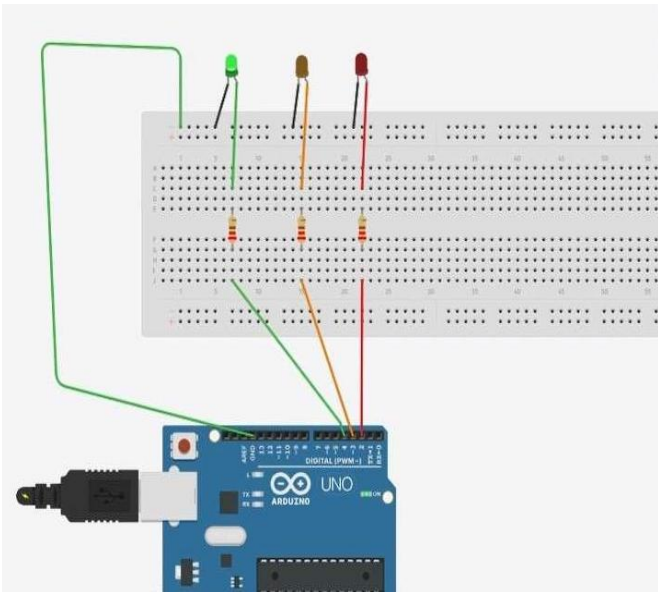

Fig 02: Simulated Circuit on Tinkercad (Green lightlight on)

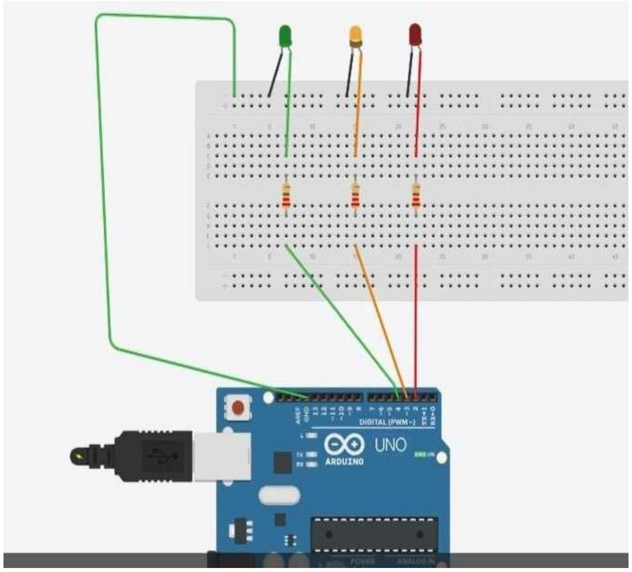

Fig 03: Simulated Circuit on Tinkercad (Orange light on)

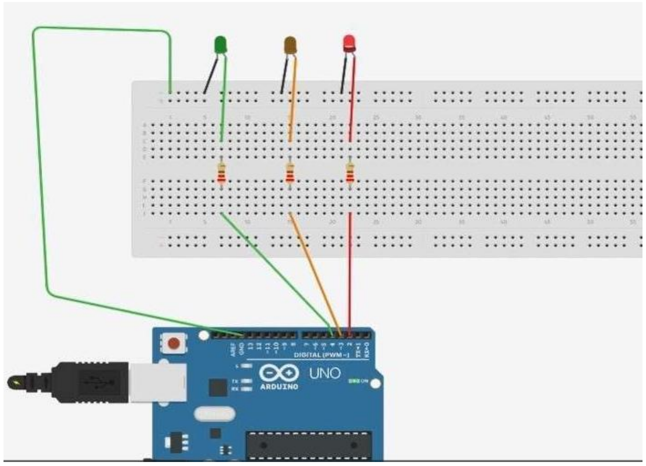

Fig 04: Simulated Circuit on Tinkercad (Red light on)

Result :

The Arduino-based traffic light system was successfully simulated using Arduino simulation software. The program was compiled and executed without any errors. Three LEDs representing **Red, Yellow, and Green** traffic lights were connected to different digital output pins of the Arduino Uno.

During the simulation, the LEDs operated in a fixed sequence:

- The **Red LED** turned ON first and remained ON for the specified time delay.
- After the red light turned OFF, the **Green LED** turned ON for its delay period.
- The **Yellow LED** then turned ON for a short duration before the system returned to the red light state.

This sequence repeated continuously, accurately representing a real-life traffic light system. The timing of each LED matched the delay values defined in the Arduino program. The simulation output confirmed that the Arduino code executed correctly and controlled the LEDs as expected.

Discussion :

The traffic light simulation demonstrates how Arduino Uno can be used to control multiple output devices using software logic. The pinMode() function was used to configure the LED pins as outputs, while the digitalWrite() function controlled the ON and OFF states of each LED.

The timing control was achieved using the delay() function, which allowed each traffic signal to remain active for a fixed duration. By adjusting the delay values in the code, the duration of each traffic light phase could be easily modified, showing the flexibility of Arduino programming.

Using simulation software made it possible to test and verify the circuit and code without physical hardware. It helped in identifying logical errors, pin configuration issues, and timing problems safely and efficiently. The simulation also provided a clear visual representation of the traffic light sequence.

Overall, the software simulation confirmed the correctness of the Arduino program and provided a reliable environment for designing and testing embedded systems before real-world implementation